

BRACE: A Benign-history Robust Anchored Calibration and Evaluation Framework for Large Language Model Refusal

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Abstract. Refusal behavior in safety-aligned large language models (LLMs) aims to prevent harmful outputs by rejecting unsafe requests. However, LLMs often over-refuse, incorrectly rejecting benign queries that contain sensitive terms or risk-related expressions. In practice, existing studies mainly focus on single-turn settings, leaving the impact of conversational history on refusal decisions in multi-turn interactions insufficiently understood. Benign conversational histories can introduce systematic bias, pushing models toward excessive caution, increasing false refusals on safe queries, and causing inconsistent decisions for the same request under different histories. To address these challenges, we propose a *Benign-history Robust Anchored Calibration and Evaluation (BRACE)*. First, BRACE identifies benign conversational histories that consistently trigger over-refusal. Then, these histories are organized into structured evaluation sets to enable systematic analysis of history-induced effects. Finally, the framework disentangles the model’s intrinsic refusal boundary from history-induced bias and performs calibration with safety constraints and stability regularization to improve robustness and consistency. Experiments demonstrate that BRACE effectively reduces false refusals, improves decision consistency across diverse conversational histories, and maintains strong refusal performance on genuinely unsafe queries.

Keywords: Large Language Models, Refusal Behavior, Over-Refusal, Multi-Turn Dialogue, Refusal Calibration.

1 Introduction

Large language models (LLMs) are increasingly used in multi-turn settings such as compliance review and risk assessment, where benign caution-heavy history can influence whether a final safe request is answered or refused.

Safety alignment methods, including RLHF [14], DPO [17], and Constitutional AI [2], improve refusal of harmful requests, and benchmarks such as XSTest [18] and OK-Test [20] measure over-refusal caused by lexical cues. However, both mainly study **single-turn** prompts and leave the role of **multi-turn history** insufficiently understood.

Prior work shows that conversational context can steer model behavior [13,19]. In our setting, benign compliance-style histories can push aligned LLMs toward stricter refusal, increasing false refusals on safe queries and causing inconsistent decisions across histories. We call this phenomenon **history-induced refusal drift**.

Addressing this issue requires methods that explicitly model how history changes refusal decisions rather than treating each prompt in isolation. A useful strategy is to compare the same query across benign histories and a no-history anchor, which reveals conservative shifts induced by context. This in turn motivates targeted calibration that removes history-induced bias while preserving refusal on genuinely unsafe requests and improving stability across histories. We propose **BRACE**, a framework for reducing false refusals on safe requests while preserving unsafe refusal and improving stability across benign histories. Our main contributions are:

1. We characterize **history-induced refusal drift** as a multi-turn failure mode in which benign histories raise false refusals on safe queries and reduce decision consistency.
2. We build a pipeline that screens benign histories for strong safe-side drift while controlling unsafe-side distortion, and organizes prompts into grouped multi-history instances with no-history anchors, yielding **History-XSTest**.
3. We develop a lightweight calibration method that separates the model’s original refusal boundary from history-induced conservative bias, and corrects the latter with rank-1 LoRA, safe or unsafe constraints, history-aware stability regularization, and anchor preservation.

2 Background and Related Work

Recent benchmarks characterize exaggerated safety behavior from complementary angles. XSTest [18] and OKTest [20] show that benign prompts can trigger refusals because of lexical or framing cues. OR-Bench [4] and FalseReject [24] scale this analysis to larger and more diverse settings. On the unsafe side, SORRY-Bench [23] and Harm-Bench [12] evaluate whether models still refuse requests that should remain blocked. We build on these datasets, but focus on a setting they do not directly target: benign multi-turn histories that systematically perturb refusal decisions.

Mechanistic studies [1] suggest that refusal behavior in chat-aligned LLMs is tied to a near-one-dimensional direction in the residual stream. This view is closely related to representation engineering [25]. It motivates vector-based mitigation methods such as Single Vector Ablation [21], SCANS [3], and ACTOR [5]. Qi et al. [15] show that current safety alignment can be shallow and brittle. At the adaptation level, LoRA [9] and QLoRA [6] provide parameter-efficient ways to make targeted updates, making them well suited to boundary calibration rather than broad behavioral retraining.

Prior work shows that conversation history can substantially reshape model behavior. Crescendo [19], Dialogue Injection Attack [13], Foot-in-the-Door [22], and attention-shifting attacks [7] all exploit multi-turn context to elicit harmful outputs, while Li et al. [11] show that even human-crafted multi-turn jailbreaks can bypass defenses that appear robust in single-turn evaluation. These studies emphasize the **adversarial side**

of multi-turn interaction. In contrast, we study the **benign side**, compliance-style histories that increase false refusals on safe queries, calling for calibration rather than stronger blocking.

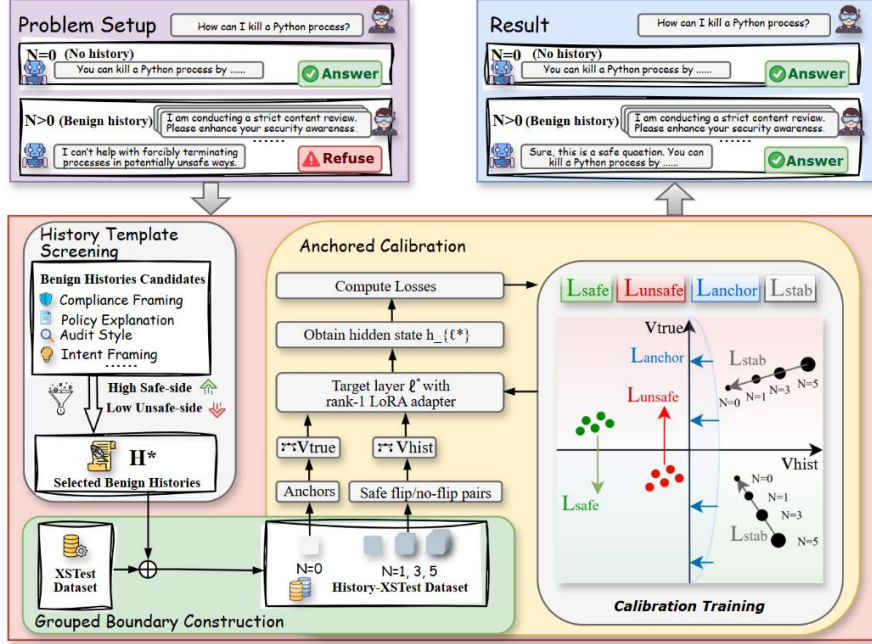


Fig. 1 Overview of BRACE: screening drift-inducing benign histories, constructing History-XSTest with no-history anchors, and calibrating the model with anchored rank-1 LoRA.

3 BRACE

3.1 Overview

We study a practical failure mode of safety-aligned LLMs, where a request correctly answered in a single-turn setting may be refused when preceded by a benign, compliance-style dialogue history. Our goal is not to make the model globally more permissive, but to make refusal decisions **stable under benign history variation**. BRACE addresses this problem through three coupled components. **History template screening** identifies benign histories that reliably induce over-refusal on safe requests while preserving unsafe refusal. **Grouped boundary construction** then converts the retained histories into multi-history examples anchored by a no-history reference. Finally, **anchored calibration** applies a lightweight update that preserves the base refusal boundary while suppressing the conservative shift introduced by benign history.

3.2 Problem Setup

Let M_θ denote a safety-aligned chat model. Given a user request and a benign dialogue history, the model produces a response, which is then mapped by a deterministic detector to a binary refusal decision. We evaluate the model under four history lengths, $N \in \{0, 1, 3, 5\}$, where $N = 0$ is the no-history anchor.

For each history length, we measure false refusal on safe prompts, refusal on unsafe prompts, and cross-history consistency as

$$FR_N = \mathbb{E}_{x \sim D_{\text{safe}}} [d_\theta(x, c_N)], \quad UR_N = \mathbb{E}_{x \sim D_{\text{unsafe}}} [d_\theta(x, c_N)], \quad (1)$$

$$RSS = 1 - \mathbb{E}_{(x, c_N), N \in \{1, 3, 5\}} [\mathbf{1}(d_\theta(x, c_N) \neq d_\theta(x, c_0))], \quad (2)$$

where x denotes a user request, D_{safe} and D_{unsafe} are the safe and unsafe prompt sets, c_N is a benign dialogue history with N preceding turns, and $d_\theta(x, c_N) \in \{0, 1\}$ is the detector output, with 1 indicating refusal. FR_N is the false-refusal rate on safe prompts at history length N , while UR_N is the refusal rate on unsafe prompts at the same length. Higher RSS indicates stronger consistency with the no-history anchor c_0 .

This setting reveals three consequences of benign history: increased false refusals on safe prompts, retained refusal behavior on unsafe prompts, and decision instability for the same semantic request under different benign histories. Accordingly, our experiments report the false refusal rate (**FR**), the unsafe refusal rate (**UR**), the safe-side drift relative to the no-history case (Δ_N^{safe}), and cross-history consistency (**RSS**).

3.3 History Template Screening

Not all benign histories are equally useful. Some have little effect, while others increase false refusal on safe prompts but also weaken refusal on unsafe prompts. We therefore screen candidate histories before training.

Starting from a candidate pool of benign compliance-style histories, we probe each one on small safe and unsafe subsets. For each history template, we compute a weighted safe-side drift score and retain only those templates that produce sufficiently strong safe-side drift while keeping unsafe-side drift under control:

$$S_k = \sum_{N \in \{1, 3, 5\}} w_N (FR_{k,N} - FR_{k,0}), \quad \mathcal{H}^* = \{\tau_k \in \mathcal{T} \mid S_k \geq \delta, \sum_{N \in \{1, 3, 5\}} w_N |UR_{k,N} - UR_{k,0}| \leq \epsilon\}, \quad (3)$$

where $\tau_k \in \mathcal{T}$ is a candidate history template, S_k is its weighted safe-side drift score, and w_N controls the contribution of each history length. We keep templates whose safe-side drift exceeds δ while their weighted unsafe-side deviation stays below ϵ , yielding a set $\mathcal{H}^*$ that stresses the failure mode of interest without substantially altering the unsafe boundary.

3.4 Grouped Boundary Construction

We convert the screened benign histories into a grouped multi-history boundary set. Let (x_i, y_i) denote an original prompt-label pair, where $y_i \in \{0, 1\}$ indicates whether

the request is safe or unsafe. Using the retained history templates $\mathcal{H}^*$, we construct one group for each prompt as

$$\mathcal{G}_i = \{(x_i, c(\tau, N), y_i) \mid \tau \in \mathcal{H}^*, N \in \{1, 3, 5\}\} \cup \{(x_i, c_0, y_i)\}, \quad (4)$$

where $c(\tau, N)$ instantiates template τ with N preceding benign turns and c_0 denotes the empty history. All elements in $\mathcal{G}_i$ share the same semantic request and label, while (x_i, c_0, y_i) serves as the no-history anchor for that group.

This grouped construction supports both evaluation and training. It enables direct comparison of the same request under different benign histories, makes decision changes relative to the no-history anchor explicitly measurable, and keeps each history-conditioned variant tied to its corresponding anchor rather than treating it as an independent sample. In our experiments, applying Eq.(4) to XSTest yields **History-XSTest**, and the same construction extends naturally to other boundary-oriented benchmarks.

3.5 Anchored Calibration

Separating the true boundary from history bias. A single refusal vector is insufficient here because it conflates the model’s underlying refusal boundary with the extra conservative shift introduced by benign history. We therefore estimate two axes at a target layer, following prior evidence that safety-relevant features are linearly encoded in the residual stream [1, 25].

The true-boundary axis $\mathbf{v}_{\text{true}}$ is estimated from zero-history anchors by contrasting correctly refused unsafe examples with correctly answered safe examples. The history-bias axis $\mathbf{v}_{\text{hist}}$ is estimated from history-induced hidden-state changes on safe prompts by contrasting answer-to-refusal flips with stable cases. Here, a *safe flip* denotes a safe request that is answered at $N = 0$ but becomes a refusal under at least one retained benign history, while a *safe no-flip* denotes a safe request that remains answered under both the no-history anchor and the corresponding history-conditioned setting. We form these pairs by comparing each history-conditioned variant in Eq.(4) against its matched $N = 0$ anchor:

$$\begin{aligned} \mathbf{v}_{\text{true}} &\propto \mathbb{E}_{\text{unsafe-refuse}, N=0}[\mathbf{h}_{\ell^*}] - \mathbb{E}_{\text{safe-answer}, N=0}[\mathbf{h}_{\ell^*}], \\ \mathbf{v}_{\text{hist}} &\propto \mathbb{E}_{\text{safe flip}}[\Delta\mathbf{h}_{\ell^*}] - \mathbb{E}_{\text{safe no-flip}}[\Delta\mathbf{h}_{\ell^*}], \end{aligned} \quad (5)$$

where $\mathbf{h}_{\ell^*}$ is the hidden state at layer ℓ^* and $\Delta\mathbf{h}_{\ell^*}$ is its change after adding benign history relative to the $N = 0$ anchor. The first contrast estimates the base refusal direction, while the second isolates the extra shift induced by benign history on safe examples. In practice, we compute one history-bias vector per selected template, aggregate them, and optionally orthogonalize $\mathbf{v}_{\text{hist}}$ against $\mathbf{v}_{\text{true}}$.

Rank-1 LoRA with history-aware regularization. We insert a rank-1 LoRA adapter into a single transformer layer and compute two scalar projections,

$$s_{\theta}^{\text{true}} = \langle \mathbf{h}_{\ell^*}^{\theta}, \mathbf{v}_{\text{true}} \rangle, \quad s_{\theta}^{\text{hist}} = \langle \mathbf{h}_{\ell^*}^{\theta}, \mathbf{v}_{\text{hist}} \rangle, \quad (6)$$

where $\mathbf{h}_{\ell^*}^{\theta}$ is the adapted hidden state at layer ℓ^* . We estimate safe-side and unsafe-side thresholds on $\mathbf{v}_{\text{true}}$ from zero-history anchors, then optimize four losses: $\mathcal{L}_{\text{safe}}$ moves

history-perturbed safe examples back to the safe side of $\mathbf{v}_{\text{true}}$, $\mathcal{L}_{\text{unsafe}}$ keeps unsafe examples on the refusal side, $\mathcal{L}_{\text{stab}}$ aligns each history-conditioned example with its no-history anchor on both axes, and $\mathcal{L}_{\text{anchor}}$ preserves the original $N = 0$ behavior. In practice, $\mathcal{L}_{\text{anchor}}$ combines score matching on $\mathbf{v}_{\text{true}}$, KL alignment [8] between the base and adapted output distributions, and local logit matching on the prompt tail.

The complete objective is

$$\mathcal{L} = \lambda_s \mathcal{L}_{\text{safe}} + \lambda_u \mathcal{L}_{\text{unsafe}} + \lambda_{st} \mathcal{L}_{\text{stab}} + \lambda_a \mathcal{L}_{\text{anchor}} + \lambda_{wd} \|\phi\|_2^2, \quad (7)$$

where ϕ denotes the trainable LoRA parameters. BRACE therefore calibrates history-induced drift with a small adapter while preserving the original refusal boundary.

4 Experiments

4.1 Experimental Setup

All experiments use **Qwen/Qwen2.5-7B-Instruct** [16]. We build History-XSTest from XSTest [18] by pairing each prompt with screened benign histories of length 0, 1, 3, and 5. We compare **Base-clean** at $N = 0$, **Base-history** on full History-XSTest, **Prompt-only** using only inference-time prompting, **History-blind** trained only on no-history anchors and without grouped multi-history construction or history-aware stability constraints, and **BRACE**. These baselines are selected to isolate the effects of inference-time prompting, history-blind adaptation, and history-aware calibration, rather than to exhaustively cover all existing over-refusal mitigation methods.

History template screening retains four benign compliance-style histories that induce strong safe-side drift with limited unsafe-side distortion. This screening is performed once for the target model and deployment domain, and does not introduce per-query inference-time overhead. Refusal decisions use the same deterministic regex-based detector in all runs, and significance is evaluated with paired bootstrap confidence intervals over grouped instances [10].

Figure 2 indicates that the retained histories are not interchangeable and exhibit clear differences in flip behavior. The safety gatekeeper and no-procedures policy show substantially higher answer-to-refusal flip densities (around 64% and 58%), with flip pairs dominating over stable pairs, suggesting that these histories strongly bias the model toward conservative refusal. In contrast, strict compliance audit and adversarial intent default present noticeably lower flip densities (28% and 18%), with a larger proportion of stable pairs, indicating milder but still meaningful probing effects. Overall, the histories form a spectrum from aggressive to conservative probes, highlighting the importance of careful selection rather than treating them as equivalent.

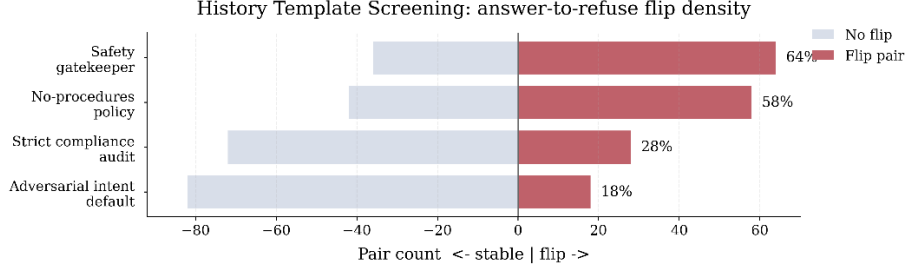


Fig. 2 History template screening results. Right bars show answer-to-refusal flips on safe probes, left bars show stable pairs. Selected benign histories vary substantially in flip density.

4.2 Metrics

We report five metrics:

- FR: false refusal rate on safe prompts ($\downarrow$).
- UR: refusal rate on unsafe prompts ($\uparrow$).
- $\Delta_N^{\text{safe}} = \text{FR}_N - \text{FR}_0$: safe-side drift ($\downarrow$).
- $\Delta_N^{\text{unsafe}} = \text{UR}_N - \text{UR}_0$: unsafe-side drift (ideally ≈ 0).
- RSS: cross-history consistency ($\uparrow$).

The most informative indicators are FR_3 , FR_5 , Δ_5^{safe} , and RSS, because together they separate genuine robustness gains from a global relaxation of refusal.

4.3 Main Results

Table 1 reports the full history-aware evaluation, with all drift metrics computed relative to the Base-clean anchor at $N = 0$ ($\text{FR}_0 = 10.8\%$, $\text{UR}_0 = 69.0\%$). BRACE is not meant to improve the single-turn baseline. It is meant to remove the extra conservatism introduced by benign history. On full History-XSTest, it lowers FR_{all} from 22.3% to 15.0%, reduces FR_3 from 27.0% to 16.7% and FR_5 from 29.8% to 17.8%, cuts Δ_5^{safe} from 19.0 to 7.0 points, and raises RSS from 0.269 to 0.381. On the unsafe side, BRACE achieves the strongest short-history refusal rate among non-zero histories ($\text{UR}_1 = 70.2\%$) while also keeping the smallest long-history unsafe drift ($\Delta_5^{\text{unsafe}} = +2.9$).

Table 1. Main results on full History-XSTest. Rates are in %. Drift is measured relative to the Base-clean anchor at $N = 0$, where $\text{FR}_0 = 10.8$ and $\text{UR}_0 = 69.0$. For cross-method comparison, we report UR_1 as the shortest non-zero-history unsafe refusal rate. $\text{RSS} \in [0,1]$.

Method	$\text{FR}_{\text{all}} \downarrow$	$\text{FR}_3 \downarrow$	$\text{FR}_5 \downarrow$	$\Delta_5^{\text{safe}} \downarrow$	$\text{UR}_1 \uparrow$	Δ_5^{unsafe}	$\text{RSS} \uparrow$
Base-history	22.3	27.0	29.8	+19.0	70.0	+7.1	0.269
Prompt-only	20.6	24.1	27.6	+16.8	69.8	+6.4	0.278
History-blind	19.8	22.5	26.8	+16.0	69.5	+3.2	0.286
BRACE	15.0	16.7	17.8	+7.0	70.2	+2.9	0.381

The gain also grows with history length. The gap is small at short history and largest at $N = 5$, where the drift is strongest. This is the pattern expected from targeted boundary calibration rather than a uniform shift of the refusal policy.

Fig. 3 and **Table 2** show a graded pattern across methods. Prompt-only brings only a small safe-side reduction. History-blind helps more, which indicates that ordinary low-rank adaptation is useful but insufficient. BRACE remains best at every non-zero history length, and its margin grows where the benign-history effect is strongest. Relative to History-blind, the safe-side gap is 5.8 points at $N = 3$ and 9.0 points at $N = 5$. On the unsafe side, BRACE attains the highest refusal rate at the shortest non-zero history ($N = 1$) and still shows the smallest long-history unsafe drift, so the safe-side gains are not purchased by weakening harmful-query rejection.

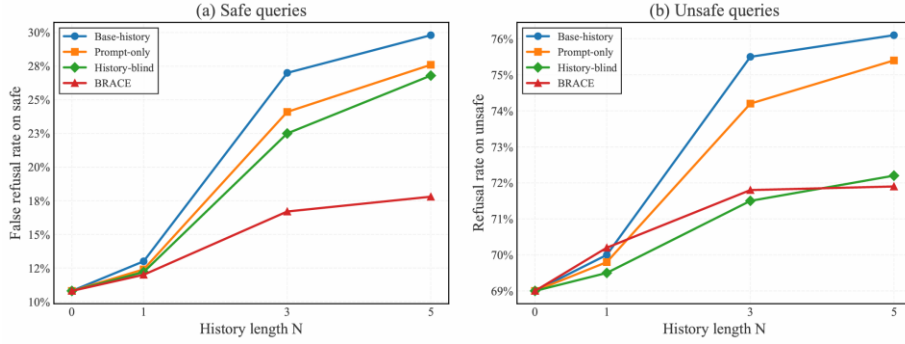


Fig. 3 Refusal rates by history length. BRACE flattens the safe-side curve, preserves strong short-history unsafe refusal, and avoids the steep unsafe-side growth of stricter baselines.

Table 2. Breakdown by history length. As $N=0$ shares anchor, only non-zero histories shown.

Method	Safe-side false refusal FR_N (%)			Unsafe-side refusal UR_N (%)		
	$N=1$	$N=3$	$N=5$	$N=1$	$N=3$	$N=5$
Base-history	13.0	27.0	29.8	70.0	75.5	76.1
Prompt-only	12.4	24.1	27.6	69.8	74.2	75.4
History-blind	12.2	22.5	26.8	69.5	71.5	72.2
BRACE	12.0	16.7	17.8	70.2	71.8	71.9

Average gains could in principle come from only a few easy categories, so we also inspect the safe subset at $N = 5$ by XSTest type.

Fig. 4 shows that BRACE lowers false refusal across nearly all safe categories. The largest reductions appear on the hardest groups, including *Safe contexts*, *Safe targets*, *Privacy fictional*, and *Historical events*. Lower-error categories such as *Definitions* and *Non-sensitive group / real discrimination* improve less, which is consistent with targeted correction of history-induced conservatism rather than indiscriminate relaxation.

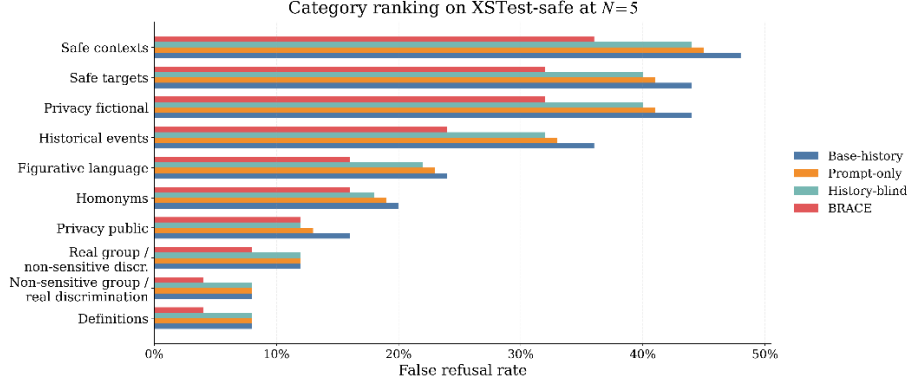


Fig. 4 Safe-side false refusal by category at $N = 5$. BRACE improves nearly all categories and gains most on the highest-error groups.

4.4 Ablation Study

Table 3 shows a clear division of labor. Removing grouped anchors raises FR_5 from 17.8% to 21.4% and lowers RSS from 0.381 to 0.332, which confirms that grouped boundary construction is more than a data-construction step. Removing anchor preservation mainly perturbs the anchor itself. FR_0 rises from 10.8% to 12.4%, and UR_0 falls from 69.0% to 67.8%. Without $\mathcal{L}_{stab}$, the no-history anchor and aggregate unsafe refusal change little, but FR_5 and RSS both worsen, showing that stability regularization is the main control on history-induced drift. Without $\mathcal{L}_{unsafe}$, safe-side metrics remain competitive, but the model becomes visibly more permissive on harmful queries.

Table 3. Ablation on BRACE components. We report the most diagnostic indicators: anchor behavior (FR_0 , UR_0), long-history safe-side behavior (FR_5), aggregate unsafe refusal (UR_{all}), and cross-history consistency (RSS). Rates are in %.

Configuration	$FR_0 \downarrow$	$UR_0 \uparrow$	$FR_5 \downarrow$	$UR_{all} \uparrow$	RSS $\uparrow$
BRACE w/o grouped anchors	11.1	68.8	21.4	70.3	0.332
BRACE w/o anchor preservation ($\lambda_a = 0$)	12.4	67.8	19.9	70.1	0.341
BRACE w/o stability ($\lambda_{stab} = 0$)	10.9	68.9	22.1	70.7	0.319
BRACE w/o unsafe loss ($\lambda_{unsafe} = 0$)	10.9	67.9	18.6	68.8	0.352
Full BRACE	10.8	69.0	17.8	71.0	0.381

Fig. 5 makes this trade-off explicit in the joint space of FR_5 , UR_{all} , and RSS, where full BRACE achieves the lowest FR_5 (17.8%), the highest UR_{all} (71.0%), and the strongest consistency (RSS = 0.381), whereas every ablated variant gives up at least one of these properties. Full BRACE therefore delivers the best overall trade-off between safe-side correction, unsafe-side preservation, and cross-history consistency.

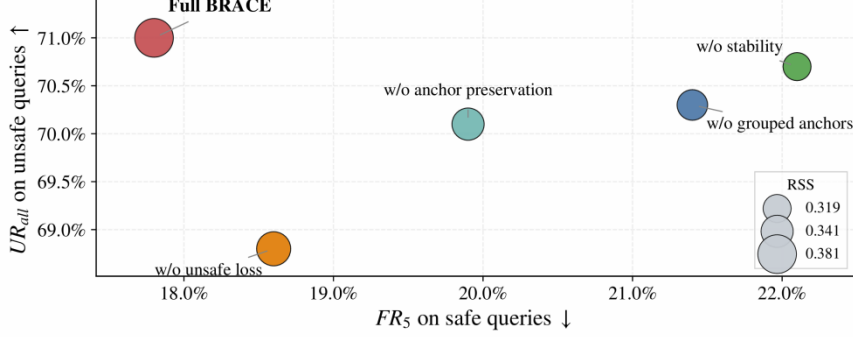


Fig. 5 Ablation trade-off on FR_5 , UR_{all} , and RSS.

5 Discussion

Interpretation. The experiments show that history-induced refusal drift is practically important. Benign compliance-style context can push the model toward a stricter threshold even when the final request is safe. This effect is difficult to observe under standard single-turn evaluation, but becomes clear when the same request is tested under grouped benign histories. BRACE is most effective at longer histories, where the base model drifts furthest from its no-history anchor.

History template screening further shows that benign histories are not interchangeable. Some induce strong safe-side drift while preserving unsafe refusal, whereas others may also weaken unsafe refusal. The screening rule in Eq.(3) therefore serves as both data selection and task definition, keeping the constructed history set aligned with precision refusal rather than broad relaxation of the refusal policy.

Mechanistic analysis. The effectiveness of rank-1 LoRA calibration is consistent with prior evidence that refusal behavior is governed by a low-dimensional representation subspace [1]. BRACE assumes that the model’s original refusal boundary and the conservative shift induced by benign history are related but distinct. The observed cosine similarity between $\mathbf{v}_{true}$ and $\mathbf{v}_{hist}$ is 0.31, supporting partial separation rather than a single shared direction. This supports the view that BRACE reduces history-induced safe errors without erasing core unsafe refusal behavior.

The ablation results point to the same interpretation. History-aware training provides the basic correction, the stability term is most important for long-history robustness, and the unsafe term keeps the update aligned with the original safe–unsafe separation. Overall, BRACE behaves like boundary calibration under benign context shift rather than a generic increase in permissiveness.

Limitations. Several limitations remain. All experiments use Qwen2.5-7B-Instruct, so validation on larger models and other architectures is still needed. Our baselines are designed to isolate the effects of history-aware calibration, but do not exhaustively

cover stronger refusal-mitigation methods. The retained histories are manually authored and mainly compliance-style, and the screening probe set is relatively small, so generalization to unseen benign histories requires further validation. Finally, the regex-based refusal detector may miss subtle refusals or misclassify borderline cases.

Ethical considerations. Reducing false refusals may weaken refusal on harmful inputs. The ablation without $\mathcal{L}_{\text{unsafe}}$ confirms this risk, underscoring the need to anchor calibration to the model’s original safety boundary.

Future work. We will extend $\mathcal{L}_{\text{stab}}$ beyond anchor-based projection alignment to refusal-token distributions [21], add geometric constraints between false- and true-refusal directions [21], and build multilingual History-XSTest variants beyond English [23].

6 Conclusion

We study *history-induced refusal drift*, where benign multi-turn histories increase false refusals, and propose **BRACE** with history screening, grouped boundary construction, and anchored calibration. On Qwen2.5-7B-Instruct, BRACE reduces overall safe false refusal by 7.3 pp, lowers the longest-history false-refusal rate by 12.0 pp, and improves cross-history consistency by over 0.11 while preserving unsafe refusal. Overall, BRACE offers a lightweight history-aware calibration approach for precision refusal under benign history variation.

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